

Homework 1

Due: Sunday 11/10/2026 23:59:00

Group submission: HW1_ **YOURNAME**.zip

Late submission policy: Late submissions will incur a **10%** deduction from the total score for **each** day late, for a maximum of 3 days. Any submission received more than 3 days after the deadline will receive a score of 0 for the assignment.

- Submit your source code + documentation (pdf) in a zip file on LMS
- Your documentation must include the **full names** of all the members in your group and their **contributions (0-100%)**. Only **ONE (1)** submission per group is required.

Choose 1 of the 2 topics:

TOPIC 1

MINESWEEPER



Minesweeper is a logic puzzle where the goal is to clear a grid of hidden mines without detonating any. It is one of the most popular singleplayer logic video games. The game features a grid of clickable tiles, with hidden "mines" dispersed throughout the board. The objective is to clear the board without detonating any mines, with help from clues about the number of neighboring mines in each field.

Rules:

- **Opening Cells:** Left-click on a square to reveal it. If it's a number, that's how many mines are in the **8** surrounding cells (horizontal, vertical, and diagonal). while blank tiles reveal nothing nearby. Use these numbers to deduce mine locations, right-clicking to place flags.
- **Winning:** The game is won when all safe, non-mine cells are uncovered.
- **Losing:** The game ends immediately if you click on a mine.

You can use **Depth-First Search (DFS)** and/or **Breath-First Search (BFS)**.

You must design a **GUI** to play this game (5x5, 9x9, ...).

What **heuristics** could be used in the game?

You must document your design and algorithm step by step in a pdf file.

Language: Python, C#, C/C++, ...

TOPIC 2:

Pathfinding in Ho Chi Minh City.

Construct a map of at least **100** points of interest in HCMC.

You define a **heuristic** that is function of vehicle density, travel speed, or flow rate... In other words, heuristic value at a point of interest is **NOT** a constant, but rather a function of different inputs (time-variant, constants,).

Based on the collected data, propose the shortest/cheapest path from Point A to Point B in HCMC using A* search. You should display your result on a map for visualization. Does the heuristic satisfy **Admissibility** and **Consistency** conditions?

Note:

You might as well use real-time data from different sources in HCMC. For example:
<https://giaothong.hochiminhcity.gov.vn/>

-The end -